

A Square-root Acceleration Control for a Fixed Geometric-response Prescription on SPARC Rotation Curves

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ABSTRACT

I compare a fixed Geometric-Medium Response (GMR) prescription for SPARC rotation curves with a globally fitted square-root excess-acceleration control. The control is $g_{\text{tot}} = g_N + A_{\text{sq}}\sqrt{g_N}$, with $A_{\text{sq}} = 9.0294883069378987 \times 10^{-6}$ optimized on the 24-galaxy calibration subset. On 175 SPARC galaxies it reaches equal-weight RMSE 40.241 km s^{-1} , slightly below the full GMR value 40.554 km s^{-1} . On the 151-galaxy non-calibration complement, the corresponding values are 41.329 and 41.870 km s^{-1} . Thus, in this implementation, the tested baryonic geometric-response invariants do not improve the main aggregate benchmark beyond a simple square-root acceleration core. The result constrains the current GMR prescription and identifies the square-root law as the empirical core that a geometric-response theory must explain.

Keywords: Galaxy rotation curves (619) — Dwarf galaxies (416) — Astrostatistics (1882)

SPARC rotation curves provide a compact setting for testing whether baryonic structure beyond the Newtonian acceleration g_N adds predictive information to fixed response laws (Lelli et al. 2016; McGaugh et al. 2016). The tested GMR prescription has no galaxy-by-galaxy retuning and contains an exact square-root response core,

$$g_\chi = A_{\text{GMR}} F(R) \sqrt{g_N}, \quad (1)$$

where $F(R) = \sin(\alpha_B) S_{\text{rad}}$ is a dimensionless baryonic-invariant factor. This note asks a narrower question than whether GMR is a complete theory: does the implemented invariant factor improve over the corresponding globally fitted square-root control on the primary SPARC benchmark?

The answer is no for the aggregate equal-weight RMSE used here. The square-root control is slightly lower than full GMR on both the full sample and the non-calibration complement (Table 1). The comparison is not a decisive galaxy-by-galaxy defeat of GMR: on the full sample, GMR has 88 lower per-galaxy RMSE values while the square-root control has 87. The point is instead one of parsimony and aggregate score. A one-parameter square-root excess control is sufficient to match or slightly exceed the main GMR aggregate performance in the present implementation.

Table 1. Primary equal-weight RMSE comparison in km s^{-1} .

Model and sample	N_{gal}	RMSE
Square-root control, full	175	40.241
Full GMR, full	175	40.554
Square-root control, non-calibration	151	41.329
Full GMR, non-calibration	151	41.870

The conclusion is therefore a constraint, not a superiority claim. The tested invariant architecture does not yet demonstrate robust predictive information beyond the square-root core, and it should not be read as a solution to dark matter or as a general replacement for halo modeling. Reduced halo profiles, especially cored empirical profiles (e.g., Burkert 1995), remain stronger flexible fitting baselines for many SPARC galaxies, and low-mass, low-surface-brightness dwarfs remain an important failure mode for this GMR implementation.

The square-root core also motivates future tests of non-scalar, compactness- or tidal-gradient-sensitive response operators in merging clusters, but no cluster-lensing claim is made in this note.

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Data Availability. The derived tables, source files, and reproduction scripts are available in the public GitHub release <https://github.com/mohammedmessaoudene-cmd/GMR-SPARC-Paper1/releases/tag/v47-rnaas-final> and archived under the Zenodo DOI <https://doi.org/10.5281/zenodo.20032301>.

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